

Midterm 1
Friday, February 25, 2022

John Doe
Name

- 6 problems total
- 75 minutes (4:00-5:15)
- Only writing instruments and nongraphing calculators are allowed
- Harder / longer problems are in the back. Allot time appropriately
- You can tear away the formula sheets in the back. You don't need to turn them in with your test; if you do tear them off, make sure to recycle them when you leave
- You can use the backs of pages if you need extra space for calculations
- If you feel you are missing any formulas, let me know

Good luck!

1. 15

2. 17

3. 16

4. 16

5. 18

6. 18

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Problem 1:

a) Calculate curl and divergence of the following vector field. State whether the field is conservative, solenoidal, both, or neither.

$$\mathbf{A} = \hat{\mathbf{R}} 5 \exp(-10R^2) / R^2$$

$$\nabla \times \mathbf{A} = 0 \quad \text{as only } A_R(R)$$

$$\begin{aligned} \nabla \cdot \mathbf{A} &= \frac{1}{R^2} \frac{d}{dR} (R^2 A_R) = \frac{1}{R^2} \frac{d}{dR} (5e^{-10R^2}) = \frac{1}{R^2} (-5 \cdot 10 \cdot 2R e^{-10R^2}) \\ &= -\frac{100}{R} e^{-10R^2} \end{aligned}$$

Conservative

b) Calculate curl and divergence of the following vector field. State whether the field is conservative, solenoidal, both, or neither.

$$\mathbf{A} = \hat{\mathbf{x}}(2x + 3yz) + \hat{\mathbf{y}}(2y + 3xz) + \hat{\mathbf{z}}(2z + 3xy)$$

$$\nabla \cdot \mathbf{A} = 2 + 2 + 2 = 6$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial_x & \partial_y & \partial_z \\ 2x+3yz & 2y+3xz & 2z+3xy \end{vmatrix} = \begin{aligned} &\hat{\mathbf{x}}(3x - 3x) \\ &- \hat{\mathbf{y}}(3y - 3y) \\ &+ \hat{\mathbf{z}}(3z - 3z) = 0 \end{aligned}$$

Conservative

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c) Some vector field $\mathbf{A}$ is related to a scalar field f as $\mathbf{A} = \nabla f$. What is the value of $\nabla \times \mathbf{A}$?

Explain briefly (1-2 sentences).

$$\nabla \times \mathbf{A} = 0 \quad \text{because} \quad \text{curl grad} \equiv 0$$

$$\nabla \times (\nabla f) = 0$$

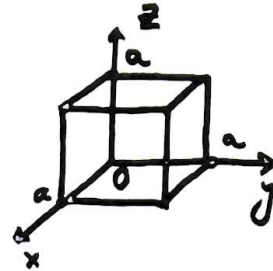
Gradient of
scalar fields
always produce
conservative fields

Problem 2: (Hint: If you know the material well, this problem will take very little time to do.)

a) Calculate the flux of the current-density vector field

$$\mathbf{J} = \hat{\mathbf{x}}(3x - 2y) + \hat{\mathbf{y}}(x - 3y) \text{ [A/m}^2\text{]}$$

through a cube of side a in the first octant of the Cartesian coordinate system, with one corner at the coordinate origin and three edges aligned with the coordinate axes, as in the picture on the right.



$$\oint \mathbf{J} \cdot d\mathbf{\hat{S}} = \int (\nabla \cdot \mathbf{J}) dV \quad \text{by divergence theorem}$$

$$\nabla \cdot \mathbf{J} = 3 - 3 = 0 \quad \text{Therefore } \underline{\oint \mathbf{J} \cdot d\mathbf{\hat{S}} = 0}$$

b) Can $\mathbf{J}$ be the current density in the static (time-independent) case? Explain briefly (1-2 sentences).

Yes, because $\nabla \cdot \mathbf{J} = 0$. By continuity equation, $\nabla \cdot \mathbf{J} + \frac{\partial \rho_v}{\partial t} = 0$ which, in the static case, reduces to $\nabla \cdot \mathbf{J} = 0$

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c) Will the total amount of charge inside this cube change with time? Explain briefly (1-2 sentences).

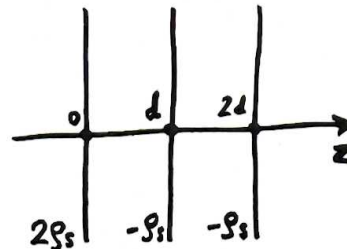
From integral form of continuity equation (formula sheet)

$$\oint \mathbf{J} \cdot d\mathbf{\hat{S}} = \int (\nabla \cdot \mathbf{J}) dV = -\frac{d}{dt} \int \rho_v dV = -\frac{d}{dt} Q_{\text{inside}}$$

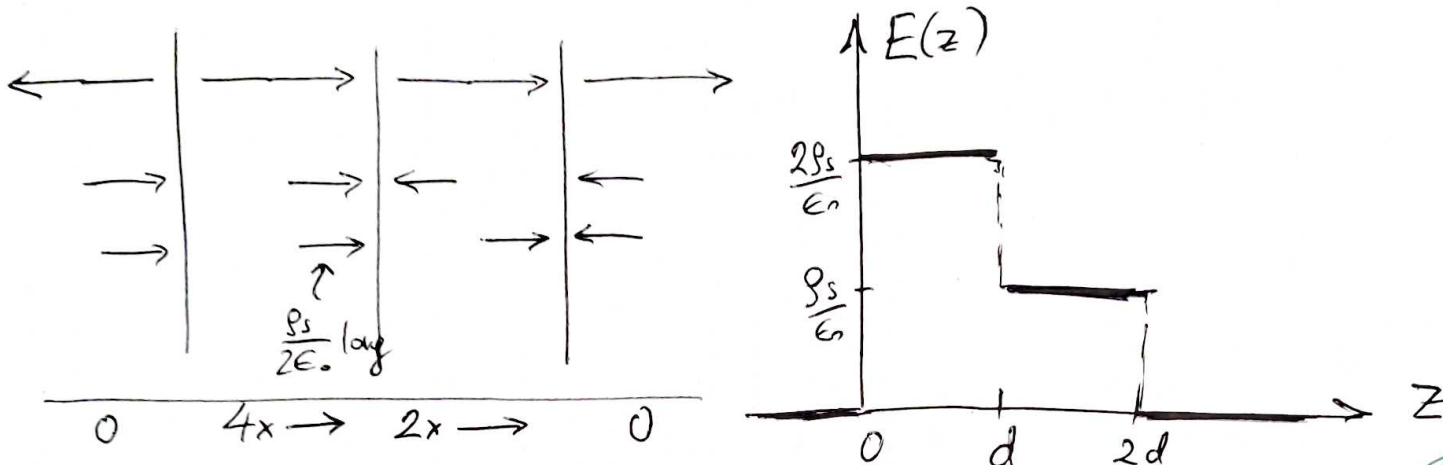
$$\begin{matrix} \parallel \\ 0 \end{matrix} \quad \text{so} \quad \frac{d}{dt} Q_{\text{inside}} = 0$$

Q_{inside} doesn't change with time

Problem 3: Three infinite parallel planes have surface charge densities $2\rho_s$, $-\rho_s$, $-\rho_s$ ($\rho_s > 0$). They are normal to the z -axis and placed at $z=0$, $z=d$, and $z=2d$, respectively, as in the figure. (Read carefully. This is NOT the same as the practice midterm!). Assume the planes are in vacuum (permittivity ϵ_0).

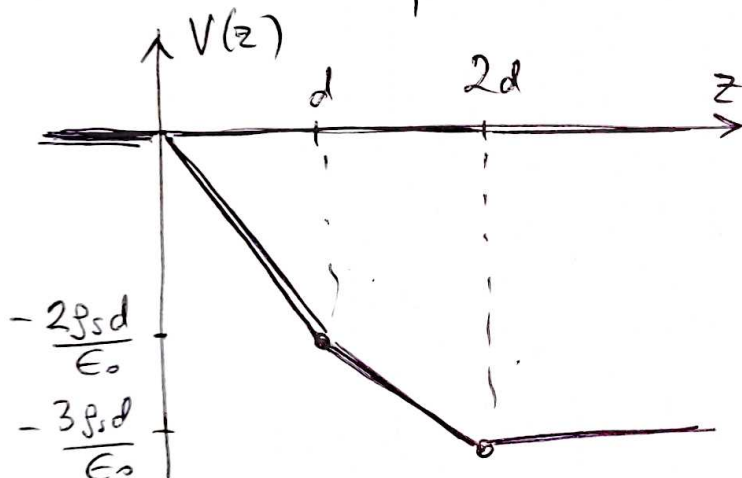


a) Find $\mathbf{E}$ everywhere in space. Graph $E(z)$ versus z .



b) Calculate and graph the electric potential $V(z)$. Assume $V(0)=0$.

Positive/negative & constant electric field $\Rightarrow$
 $\Rightarrow$ linear potential with negative/positive slope



$$\begin{aligned}
 z \leq 0 \quad V(z) &= V(0) = 0 \\
 0 < z < d \quad V(z) &= -\int_0^z \frac{2\rho_s}{\epsilon_0} dz \\
 &= -\frac{2\rho_s}{\epsilon_0} z \\
 d < z < 2d \quad V(z) &= V(d) - \int_d^z \frac{\rho_s}{\epsilon_0} dz \\
 &= -\frac{2\rho_s}{\epsilon_0} d - \frac{\rho_s}{\epsilon_0} (z - d) \\
 z \geq 2d \quad V(z) &= V(2d) = -\frac{3\rho_s}{\epsilon_0} d
 \end{aligned}$$

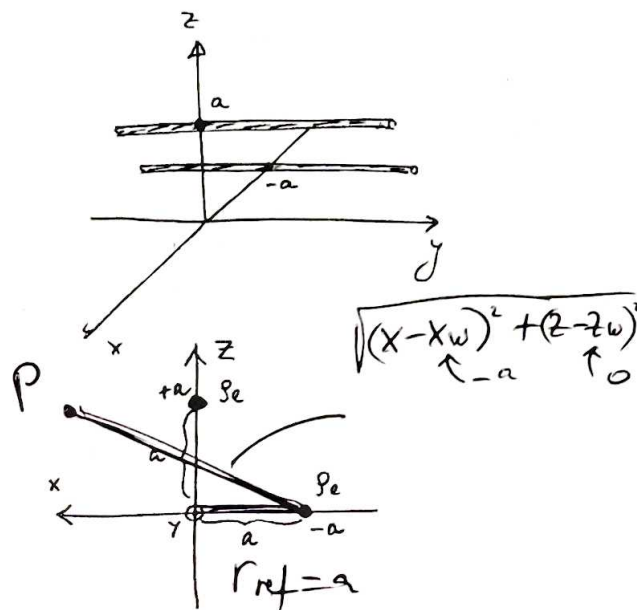
Problem 4: As discussed in class and in homework, the potential from an infinite thin wire with line charge density ρ_l which is aligned with the z -axis, can be written as

$$V(r) = \frac{\rho_l}{2\pi\epsilon} \ln\left(\frac{r_{ref}}{r}\right)$$

where r_{ref} is normal distance to the reference point.

Generalize this result and use the principle of superposition to obtain the potential arising from two infinitely long wires, both parallel to the y -axis, one located at $x = -a$ and $z = 0$, the other at $x = 0$ and $z = a$ (see picture). The two wires have the same line charge density ρ_l . The material throughout is vacuum.

Write down the potential at a point of observation P with arbitrary (x, y, z) coordinates. The reference point is the coordinate origin $(0,0,0)$.



The critical part here is writing down normal distance from point of observation to wire & reference point to wire, for each wire.

Wire 1: $r_{ref} = a$ $r = \sqrt{(x+a)^2 + z^2}$

Wire 2: $r_{ref} = a$ $r = \sqrt{x^2 + (z-a)^2}$

$$\begin{aligned} V(x, y, z) &= \frac{\rho_l}{2\pi\epsilon_0} \ln\left(\frac{a}{\sqrt{(x+a)^2 + z^2}}\right) + \frac{\rho_l}{2\pi\epsilon_0} \ln\left(\frac{a}{\sqrt{x^2 + (z-a)^2}}\right) \\ &= \frac{\rho_l}{2\pi\epsilon_0} \ln\left(\frac{a^2}{\sqrt{(x+a)^2 + z^2} \sqrt{x^2 + (z-a)^2}}\right) \\ &= \frac{\rho_l}{4\pi\epsilon_0} \ln\left(\frac{a^4}{[(x+a)^2 + z^2][x^2 + (z-a)^2]}\right) \end{aligned}$$

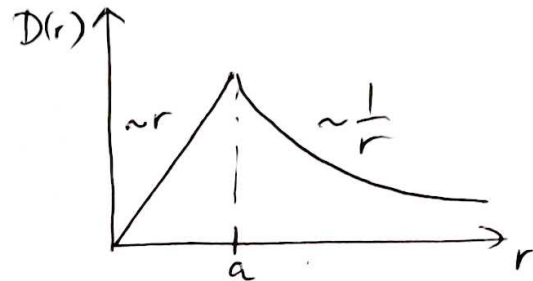
Problem 5: An infinitely long cylinder of radius a is charged with a uniform volume charge density ρ_{v0} . There is no charge outside the cylinder. Assume the permittivity of the cylinder is $\epsilon_r \epsilon_0$ ($\epsilon_r > 1$), whereas that of the surrounding medium is ϵ_0 .



a) Use Gauss's law to calculate $\mathbf{D}$ everywhere in space. Graph $D(r)$ vs r .

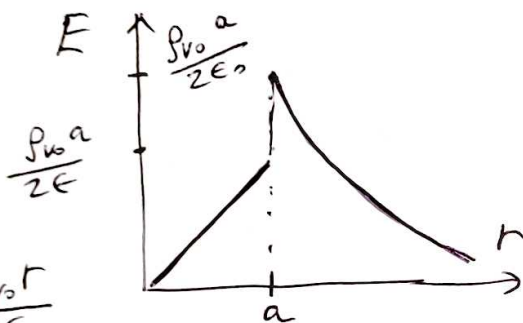
$$\begin{aligned} r < a \quad 2\pi r h D(r) &= \rho_{v0} \cdot 2\pi r^2 h \Rightarrow D(r) = \frac{\rho_{v0} r}{2} \\ r \geq a \quad 2\pi r h D(r) &= \rho_{v0} \pi a^2 h \Rightarrow D(r) = \rho_{v0} \frac{a^2}{2r} \end{aligned}$$

$$D(r) = \begin{cases} \frac{\rho_{v0} r}{2}, & r < a \\ \frac{\rho_{v0} a^2}{2r}, & r \geq a \end{cases}$$

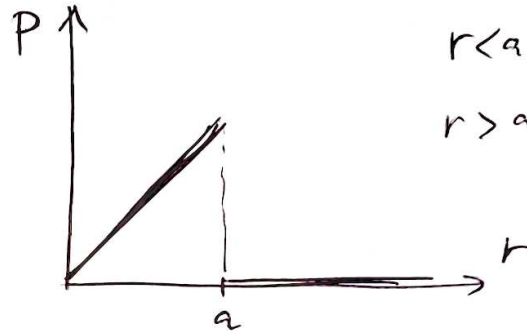


b) Calculate and graph $E(r)$ vs r everywhere. Calculate and graph $P(r)$ vs r everywhere.

As dielectric permittivity is ϵ for $r < a$ & ϵ_0 for $r > a$, E will have a discontinuity



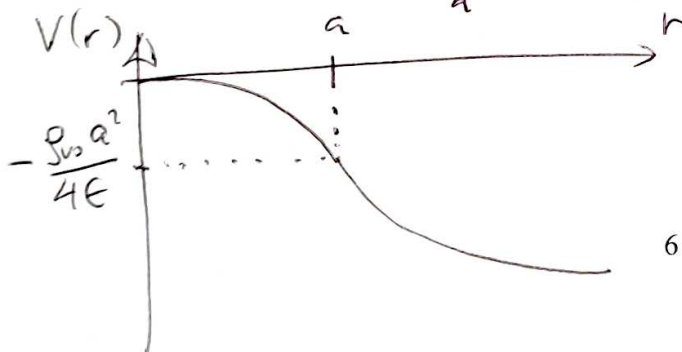
$$\begin{aligned} r < a: & \frac{\rho_{v0} r}{2\epsilon} \\ r > a: & \frac{\rho_{v0} a^2}{2\epsilon_0 r} \end{aligned}$$



$$\begin{aligned} r < a: & \frac{\epsilon_r - 1}{\epsilon_r} \frac{\rho_{v0} r}{2} \\ r > a: & 0 \end{aligned}$$

c) Calculate and graph the electric potential $V(r)$, assuming it is zero on the cylinder axis.

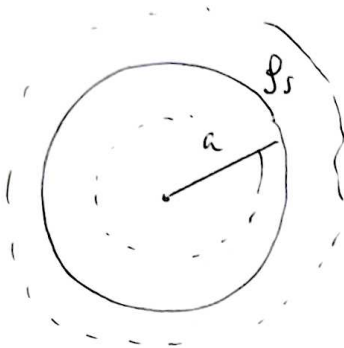
$$\begin{aligned} r < a: \quad V(r) &= - \int_0^r E(r) dr = - \int_0^r \frac{\rho_{v0} r}{2\epsilon} dr = - \frac{\rho_{v0} r^2}{4\epsilon} \\ r > a: \quad V(r) &= V(a) - \int_a^r E(r) dr = - \frac{\rho_{v0} a^2}{4\epsilon} - \int_a^r \frac{\rho_{v0} a^2}{2\epsilon_0 r} dr = - \frac{\rho_{v0} a^2}{4\epsilon} - \frac{\rho_{v0} a^2}{2\epsilon_0} \ln\left(\frac{r}{a}\right) \end{aligned}$$



Problem 6:

An infinitely thin spherical shell of radius a has a uniform surface charge density $\rho_s > 0$. Assume the surrounding medium is air (permittivity ϵ_0).

a) Use Gauss's law to calculate the electric field $\mathbf{E}$ everywhere in space. Graph $E(R)$ vs R .

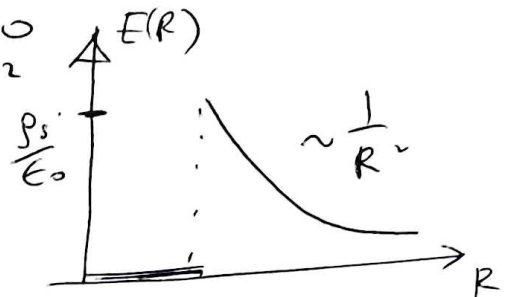


$$R < a : D(R) = 0 \Rightarrow E(R) = 0$$

$$R > a : D(R) \cdot 4\pi R^2 = \rho_s \cdot 4\pi a^2$$

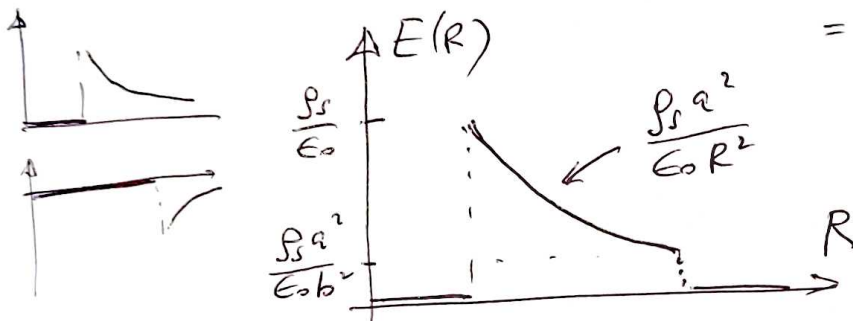
$$D(R) = \frac{\rho_s a^2}{R^2}$$

$$E(R) = \frac{\rho_s a^2}{\epsilon_0 R^2}$$



b) Another thin spherical shell, with the surface charge density $-\rho_s/4$ and radius $b=2a$, is placed concentrically around the shell from part a). Based on your result from a) and using the principle of superposition, calculate and graph the total electric field $E(R)$ vs R everywhere in space for the two spheres together.

The other shell alone has $Q(b) = -\frac{\rho_s}{4} \cdot 4\pi 4a^2 = -\rho_s 4\pi a^2$
 $= -Q(a)$ So at $R > b$ they will cancel each other



c) Calculate the potential difference between the inner and outer shells, i.e., $V(a) - V(b)$.

$$\begin{aligned} V(a) - V(b) &= - \int_b^a E(R) dR = \int_a^b E(R) dR \\ &= \int_a^b \frac{\rho_s a^2}{\epsilon_0 R^2} dR = \frac{\rho_s a^2}{\epsilon_0} \left(-\frac{1}{R} \right) \Big|_a^b \\ &= \frac{\rho_s a^2}{\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right) = \frac{\rho_s a (b-a)}{\epsilon_0 b} \\ &= \frac{\rho_s a}{2\epsilon_0} \end{aligned}$$

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